10.2 - Simple Interest

Problem Solving, Math 1000

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1 Definition

Interest is the price someone pays for the use of someone else's money. Basically, interest is the price of "renting" money. In general, we phrase this in terms of the principal, denoted P, the interest rate, denoted r, and the term of the loan, denoted t. The principal is the sum of money that the lender gives the borrower, the interest rate is the cost of "renting" the money, and the term is the length of time the borrower is borrowing the money. We often also talk about a repayment schedule, which is the manner in which the loan is repaid. If the loan is repaid all at once, we call this a single payment loan. If it's paid in chunks (say, once a month over the term of the loan), we call it an installment loan.

2 Definition

When a loan is based on simple interest, the interest rate is applied only to the principal P. This is to say, the interest paid on the loan per unit of time is constant (basically like a rental apartment, where the rent is the same every month).

3 Definition

When a loan is based on compound interest, the interest accumulates. That is to say, you first pay interest on the principal, then on the principal plus the interest it has accrued. For compound interest loans, making payments during the life of the loan decreases the total interest paid. This is why most loans with compound interest, like car loans or home mortgages, require installment payments.

4 Theorem (The Simple Interest Formula)

Let r, the interest rate of a loan for a given time period (generally, the annual percentage rate, or yearly interest rate), be a percentage expressed in decimal form. Further, let P be the principal and let t be the term of the loan. Then, the interest on the loan is rP per year over the term of the loan. So, if the term of the loan is t years, the total interest, denoted I, is given by $I = trP$. We call this formula the simple interest formula. All together, for a loan like this, the borrower will end up paying back $P + trP$ (that is, they pay back what they borrowed, plus the total interest).

4.1 Example

Many local and State governments issue bonds, as does the Federal government. Bonds last for some duration (3 years, 5 years, etc.), and often pay some amount of interest when the bond reaches maturity, or when the term of bond ends. When buying and selling bonds on the market, they cost a certain amount called the face value or par value, which is the amount you get back from the issuing agency when the bond reaches maturity (in the real world, the price bonds are actually sold at is often based on a variety of other factors as well, but we will disregard that here).

So, if you buy a five-year bond with face value \$1000 and an annual percentage rate of 3.25%, you can find the original price of the bond by solving for P in the formula:

$$F = P + trP$$

In this example, our F is \$1000, our r is $\frac{3.25}{100} = 0.0325$, and our t is 5. So, solving:

$$\begin{aligned}
F &= P + trP \\
1000 &= P + (5)(0.0325)P \\
1000 &= P\big(1 + (5)(0.0325)\big) \\
P &= \frac{1000}{1 + (5)(0.0325)} \\
P &= \$860.22 \ \ \text{(rounded to the nearest penny)}
\end{aligned}$$